



Light output response of KamLAND liquid scintillator for protons and ^{12}C nuclei

S. Yoshida^{a,*}, T. Ebihara^a, T. Yano^a, A. Kozlov^{a,1}, T. Kishimoto^b, I. Ogawa^b, R. Hazama^{b,2}, S. Umehara^b, K. Mukaida^b, K. Ichihara^b, Y. Hirano^{b,3}, I. Murata^c, J. Datemichi^c, H. Sugimoto^c

^a Research Center for Neutrino Science, Tohoku University, Aramaki aza-aoba, Aoba-ku, Sendai 980-8578, Japan

^b Department of Physics, Osaka University, Machikaneyama-cho 1-1, Toyonaka, Osaka 560-0043, Japan

^c Department of Nuclear Engineering, Osaka University, Yamadaoka 2-1, Suita, Osaka 565-0871, Japan

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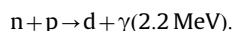
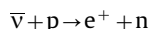
ABSTRACT

The light output responses for protons and carbon nuclei in the KamLAND liquid scintillator were precisely measured using a monochromatic neutron beam. The observed response for proton recoils is well described by Birks formula with the higher-order correction, for the recoil energy from 424 keV to 10.5 MeV. The response for carbon recoils is also well described by modifying the Birks formula with the nuclear energy loss terms, for the measured recoil energy from 171 keV to 2.2 MeV. The obtained light output responses were used to estimate the background energy spectrum of the $^{13}\text{C}(\alpha, n)^{16}\text{O}$ reaction in the KamLAND. The systematic uncertainty of the energy scale in the expected background spectrum was further improved with the measured light output responses.

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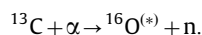
1. Introduction

The Kamioka Liquid scintillator Anti-Neutrino Detector (KamLAND) observes electron anti-neutrinos emitted from reactors [1–3], the Earth [4,5], and the Sun [6] via the inverse β -decay reaction:



The reactions can be identified by observing the correlated signals from the prompt positron and the delayed 2.2 MeV γ -ray induced by the neutron capture reaction. This distinctive signature helps to reduce the number of accidental background events and thus allows the highly sensitive electron anti-neutrino observations.

The dominant background in the most recent KamLAND analysis [3] is caused by the α capture reactions



* Corresponding author. Tel.: +81 22 795 6727; fax: +81 22 795 6728.

E-mail address: sei@awa.tohoku.ac.jp (S. Yoshida).

¹ Present address: Institute for the Physics and Mathematics of the Universe, The University of Tokyo, 5-1-5 Kashiwanoha, Kashiwa 277-8568, Japan.

² Present address: Graduate School of Engineering, Hiroshima University, 1-4-1 Kagamiyama, Higashi-Hiroshima 739-8527, Japan.

³ Present address: Department of Investigative Radiology, National Cardiovascular Center, Fujishiro-dai 5-7-1, Suita 565-8565, Japan.

The produced neutron scatters off protons and carbon nuclei in the detector forming the prompt signal. After being thermalized, the neutron is captured on a proton or ^{12}C nucleus with a mean capture time of $\sim 200 \mu\text{s}$, resulting in a delayed signal identical to that of the anti-neutrino signature. While the ^{13}C abundance is only 1.1% of the carbon isotopes in the KamLAND liquid scintillator (LS), the rate of $^{13}\text{C}(\alpha, n)^{16}\text{O}$ reaction is significant owing to a high α -decay rate of ^{210}Po .⁴

The expected background spectrum from $^{13}\text{C}(\alpha, n)^{16}\text{O}$ reactions was estimated by the number of ^{210}Po decays, the ionization energy loss of α particles in the LS [7], the reaction cross-section [8], the angular distribution of emitted neutrons [9], and the KamLAND detector responses. The detector responses include not only the energy resolution and the detection efficiency, but also the light output responses for nuclear recoils.

The light output responses in organic scintillators are non-linear in the energy deposited by ionizing particles, and depend on particle types and their energies. A small fraction of ionization energy is converted to scintillation photons via fluorescent de-excitations (quenching effect) [10]. This is particularly true for low energy particles and the particles heavier than an electron.

⁴ The ^{210}Po is produced by the decay of ^{210}Pb . The ^{210}Pb resulted from the decay of ^{222}Rn contamination which was introduced into the detector during the LS filling process. The ^{210}Pb has accumulated due to its long half-life of 22.3 yr, and is distributed within detector.

Several authors have investigated the light output responses in various kinds of LS [11] and proposed several models of response functions [12,13]. Birks suggested using the response function [12] which relates the differential light output (dL/dx) to the energy loss (dE/dx) for a given type of charged particles,

$$\frac{dL}{dx} = \frac{S\left(\frac{dE}{dx}\right)}{1 + kB\left(\frac{dE}{dx}\right)} \quad (1)$$

where S is the absolute scintillation efficiency and kB is referred to as “Birks constant”. This Birks formula well describes the quenching phenomena observed for heavier charged particles, and it successfully reproduces the experimental data of the light output responses.

In a previous analysis of the KamLAND data [2], the light output response for protons was estimated using the Birks formula. The Birks constant for protons was assumed to be the same as for α -particles ($kB=0.0115 \text{ g/cm}^2/\text{MeV}$); the light output response for carbon nuclei was not considered. Nevertheless, the uncertainty of the light output responses did not contribute a significant error to the result of the reactor anti-neutrino analysis because most of the $^{13}\text{C}(\alpha,n)^{16}\text{O}$ events were below the energy threshold of 2.6 MeV. In later KamLAND analyses focused on the observation of geo-neutrino [4]⁵ and the search for neutron disappearance [14], the light output responses for proton recoils were further improved. The validity of the response model was confirmed by comparing the energy spectrum of an Am-Be neutron calibration source with the result of Monte Carlo simulations [14]. Despite the obvious improvement, the remaining energy scale uncertainty of this method resulted in a large systematic error in the anti-neutrino analysis in the energy region where the $^{13}\text{C}(\alpha,n)^{16}\text{O}$ contribution is significant.

The goal of this study is to estimate the precise background energy spectrum of the $^{13}\text{C}(\alpha,n)^{16}\text{O}$ reaction to improve the accuracy of anti-neutrino measurements in KamLAND. The light output response, represented by the Birks constant kB in Eq. (1), depends on the LS composition as well as particle types and energy. We therefore investigate the light output response in the KamLAND LS for protons and carbon nuclei under conditions similar to those in KamLAND itself. As the kinetic energy of the neutrons produced in the $^{13}\text{C}(\alpha,n)^{16}\text{O}$ reaction ranges from 2.2 to 7.5 MeV, the energy transferred to the scattered protons and carbon nuclei can extend up to several MeV. This work presents the precise measurements of the light output response for protons and carbon nuclei in the KamLAND LS within the energy region of several MeV using a 14.1 MeV monochromatic neutron beam.

2. Experimental procedure

Measurement of the light output responses for nuclear recoils is performed using elastic scattering of monochromatic neutrons onto the target LS. A schematic view of the full experimental setup is shown in Fig. 1. A collimated 14.1 MeV neutron beam, shown in Fig. 1(b), is produced by $^3\text{H}(\text{d},n)^4\text{He}$ reactions. These neutrons are scattered by nuclei in the target LS whose light output responses are under study. The scattered neutrons are detected by an array of seven liquid scintillation detectors (BC501A in Fig. 1(c) and (d)) surrounding the target detector. More details on the experimental setup are described in the subsequent sections.

The recoil energy E_R deposited in the target detector by elastic scattering is calculated from the kinematics of the coincidence

measurement between the target LS and the BC501A detectors,

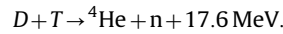
$$E_R = \left[1 - \left(\frac{m_n \cos \theta - \sqrt{m_n^2 - m_N^2 \sin^2 \theta}}{m_n + m_N} \right)^2 \right] E_n \quad (2)$$

where E_n is the incident neutron energy, m_n and m_N are the masses of the neutron and the recoiled nucleus (proton or ^{12}C), respectively, and θ is the neutron scattering angle relative to the neutron beam axis in the laboratory frame. The recoil energy E_R can be accurately calculated since the incident neutron energy E_n is monochromatic, and the scattering angle θ is fixed by placing the BC501A detectors at known angles.

The energy of incident neutrons exceeds the threshold of inelastic reactions with the target nuclei. The type of interaction (elastic or inelastic) can be determined by measuring the time-of-flight (TOF) of scattered neutrons between the target LS and the BC501A detectors. The measurement of TOF enables selection of single elastic scattering events by rejecting inelastic and multiple scattering events.

2.1. Neutron beam

The Intense 14 MeV Neutron Source Facility OKTAVIAN [15] at Osaka University was used as a neutron generator. The facility generated $\sim 10^9$ neutrons/s by bombarding a tritium implanted titanium (TiT) target with the 350 keV deuteron beam from Cockcroft–Walton accelerator in DC mode. The 14.1 MeV neutrons were emitted via the reaction



The spread of the neutron kinetic energy was estimated to be around 50 keV. The main cause of this spread was energy loss of the deuteron beam in the TiT target.

The isotropically emitted neutrons were well collimated. The neutron source (TiT target) was located in the irradiation hall surrounded by a 1 m thick concrete wall (Fig. 1(a)). The neutrons emitted at an angle of 90° relative to the deuteron beam axis were transported to the target through a 10 cm diameter cylindrical hole in the concrete wall. A lead collimator (20 cm long, 5 cm inner diameter, and 10 cm outer diameter) was inserted into the hole to shield γ -rays originating from the TiT target (Fig. 1(a) and (b)).

2.2. Target LS

The KamLAND LS was used as a target detector. The LS was a mixture of 80 vol% of normal-dodecane, 20 vol% of 1,2,4-trimethylbenzene (i.e. pseudocumene, PC), and 1.36 g/l of 2,5-diphenyloxazole (PPO) as a fluor.

A Pyrex glass cylinder (1 mm thick, 50 mm diameter, and 48 mm long) was used as a container for the target LS. The container was wrapped with PTFE reflector tape (BICRON; BC630). Two photomultiplier tubes (PMT) (2 in. diameter, Hamamatsu; H1161) were coupled to both ends of the cylinder via optical grease (BICRON; BC642).

The target LS was located in an experimental hall (Fig. 1(a)) which was separated from the TiT target room by the concrete wall. The target LS was placed about 50 cm away from the exit of the wall collimator.

In general, the light output response of LS depends on its chemical composition as well as its temperature, oxygen content, and water content [16,17]. The temperature of the target LS was measured with a K-thermocouple and kept at $9.0 \pm 1.0^\circ\text{C}$ throughout the course of the experiment. The oxygen content in the target LS was set at 13 ± 3 and 78 ppm (the maximum

⁵ The energy threshold of geo-neutrino analysis was set down to 0.9 MeV.

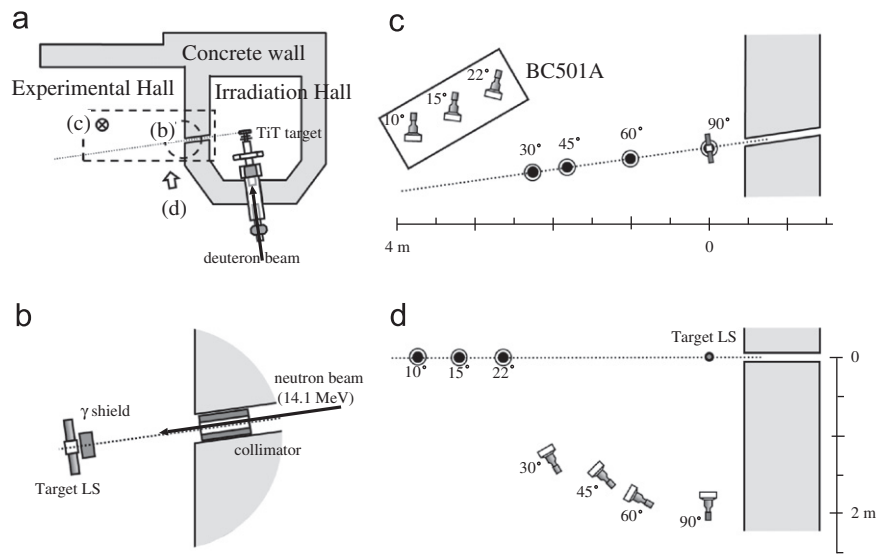


Fig. 1. (a) Schematic diagram of the experimental facility. (b) Zoomed view around the target LS. (c,d) The BC501A detector setup in relation to the target and the neutron beam axis. The three forward BC501A detectors were located in a horizontal scattering plane (c), and another four detectors in a vertical scattering plane (d) under the neutron beam line.

Table 1
The measured quenching factors for protons and carbon nuclei in the KamLAND LS.

Angle (deg)	Distance (m)	Proton recoil		Carbon recoil	
		Recoil energy (keV)	Quenching factor	Recoil energy (keV)	Quenching factor
10	4.00	$423.5^{+6.2}_{-6.1}$	0.0690 ± 0.00138	35.8 ± 0.6	
15	3.50	$940.7^{+10.5}_{-10.4}$	0.1173 ± 0.00147	80.2 ± 0.9	$\leq 5.41 \times 10^{-2}$
22	3.00	$1971. \pm 18.$	0.2075 ± 0.00214	170.8 ± 1.6	$2.91 \pm 0.25 \times 10^{-2}$
30	2.75	$3510. \pm 46.$	0.2987 ± 0.00407	312.6 ± 4.3	$1.68 \pm 0.14 \times 10^{-2}$
45	2.71	$7088. \pm 57.$	0.4430 ± 0.00396	674.6 ± 6.3	$9.00 \pm 0.63 \times 10^{-3}$
60	2.14	10530^{+66}_{-67}	0.5199 ± 0.00392	$1133. \pm 10.$	$6.54 \pm 0.42 \times 10^{-3}$
90	1.76			$2176. \pm 13.$	$4.38 \pm 0.22 \times 10^{-3}$

Scattering angle, distance from the target LS, recoil energies for proton and carbon for each BC501A detector are also listed.

dissolved level) in the two independent sets of measurements, and no significant difference was observed in the light output responses. Thus the oxygen contents did not affect the light output response, while the oxygen content in the KamLAND LS has been measured to be less than 3 ppm. The target LS was taken from the center of the KamLAND detector using a specialized sampling device [18]. Prior to the measurement, the LS sample was kept in an air tight stainless steel container, and was later carefully transferred to the detector cylinder under a nitrogen environment. As a result of these precautions the water content in the target LS is expected to be very close to that in KamLAND, ~ 20 ppm. Thus the measurement was performed under conditions very similar to those in the real KamLAND detector.

2.3. Scattered neutron detectors BC501A

The scattered neutrons were detected by an array of seven liquid scintillators (BICRON; BC501A) placed at the angles of 10° , 15° , 22° , 30° , 45° , 60° , and 90° relative to the neutron beam axis. The recoil energy deposited in the target LS was determined by the location (scattering angle in Eq. (2)) of the BC501A detectors. Simultaneous detection of the light output within the whole range of recoil energies allowed us to minimize the systematic errors of the measurement.

An aluminum cylinder (220 mm diameter and 80 mm long) was used as a container for the BC501A detector. The light from the BC501A was detected by 5 in. PMTs (Hamamatsu; R1205).

The deployment of the BC501A detectors was not isotropic. At the forward scattering angles the BC501A detectors were placed further from the target (see Table 1). The axes of the BC501A detectors were oriented perpendicular to the flight direction of the scattered neutrons⁶ (Fig. 1(c) and (d)). These specific detector setups allowed us to reduce the spread of the recoil energies. The effectiveness of the background rejection capability using the TOF information depends on the distance between the target and BC501A detectors. The energy of the scattered neutrons increased as the scattering angle decreased. The oval-shaped placement for the detectors located at the forward angles also helped to reduce the background from γ -rays and inelastic/multiple neutron scattering in the target with the TOF information.

The ^{22}Na 511 and 1274 keV γ -rays were used to set the dynamic range of the BC501A detectors at 10–15 MeV (electron-equivalent). The energy thresholds of the BC501A detectors were set individually around 100 keV. The calibration of TOF between the target and the

⁶ The BC501A detector placed at 90° was set in front of the horizontal plane, since the event rate of proton recoils was extremely small.

BC510A detectors was performed using the annihilation γ -rays of ^{22}Na as well.

2.4. Energy calibration of the target LS

Light output responses for nuclear recoils are usually expressed by the quenching factor (QF), which is defined as the ratio of produced scintillation light amount by nuclear recoils to that by electron recoils with the same energy. The detector is calibrated with a standard γ -ray source, which yields the light output in terms of an electron-equivalent energy.

An electron-equivalent energy scale of the target detector was calibrated with the backward Compton scattering γ -ray of ^{137}Cs . The experimental setup for the calibration is shown in Fig. 2(a). A NaI(Tl) detector (2 in. in diameter and 2 in. in length) was placed at the opposite side of the target LS with respect to the ^{137}Cs source.

Coincidence events between the target LS and the NaI(Tl) detector form a peak in the pulse height spectrum of the target LS at energy 477.3 keV, as shown in Fig. 2(b). A typical energy resolution was measured to be 13.9% at 477.3 keV. This backward-scattering calibration enabled us to determine precisely the energy scale, compared with the calibration by the Compton edge.

To check the shift of the energy scale, the energy calibration was performed before and after the nuclear recoil measurement. By comparing these energy spectra with each other, the shift of the energy scale was found to be less than 0.3%.

Two different modes of PMT gains were employed for the target detector. The QF of carbon recoils in the LS was assumed to be extremely small, compared with that of proton recoils. The lower gain mode was used to detect proton recoil events, where the energy threshold of the target LS was set to ~ 30 keV and the dynamic range up to 12 MeV. The carbon recoil events were measured under

the higher PMT gain. In this case, the PMT signals were discriminated at the level below single photoelectron. The trigger of the target LS was made by coincidence of two PMT's. The observed photoelectron yield of the target LS was 0.6 p.e./keV, hence the hardware energy threshold was expected to be ~ 3.5 keV. The energy scales were individually calibrated for both PMT gains.

3. Analysis and results

3.1. Event selection

The light output response for nuclear recoils was obtained from single elastic scattering events in the target LS. The trigger for such events was a coincidence signal between the target LS and the one of BC501 detectors. The criteria to select the candidate events were as follows.

- The signal detected by BC501A was induced by a neutron, not a γ -ray.
- TOF was consistent with the neutron energy expected from a single elastic scattering event.

Scattered neutrons were detected by BC501A mainly via proton recoils. It is well known that organic scintillators can separate neutron scattering (proton recoil) events from γ -ray (electron recoil) events by pulse shape discrimination (PSD) technique. The signals of neutron events have a longer tail than those of γ -ray events for the same energy deposits in the LS. Therefore, using the ratio of the total charge to the charge in the prompt part of the signal, the neutron events can be clearly separated from the γ -ray events.

The signals from BC501A detectors were sent to two charge ADCs, integrating the charge over different time intervals within 400 ns (total) and 80 ns of a prompt part (prompt). Fig. 3(a) shows the two-dimensional plot between the total and the prompt charges for BC501A. A PSD parameter was defined as

$$\text{PSD parameter} = \frac{\text{Total} - (a \times \text{prompt})}{\text{Total} + (a \times \text{prompt})} \quad (3)$$

where a is the ratio of ADC scales. Fig. 3(b) shows the distribution of the PSD parameter for the coincidence events with energy above 150 keV (electron-equivalent) in BC501A. The neutron events were sufficiently discriminated from γ -ray events.

Fig. 3(c) shows TOF spectra for the coincidence events with and without the PSD selection in BC501A. While a flat background was distributed over a wide range of the TOF spectrum due to accidental coincidences, these were strongly reduced by the PSD selection. There were three peaks in the TOF spectrum without PSD selection. The primary arriving events were from γ -rays, and the delayed two peaks from neutrons (carbon and proton recoils in the target LS). By applying TOF selection, the backgrounds from multiple/inelastic neutron scattering events in the target detector were expected to be eliminated efficiently.

3.2. Recoil energy spectrum and quenching factor

The energy spectra for proton recoils in the target LS are shown in Fig. 4(a)–(f) for each scattering angle. The horizontal axes of the recoil spectra are represented with the calibrated electron-equivalent energy scale. The spectrum of coincidence events with the BC501A placed at 90° is not shown, since there are almost no events induced by a single proton recoil in the target LS. (The kinetic energy of scattered neutrons is nearly zero at 90° .)

The obtained recoil spectra were free from backgrounds. The possible backgrounds from (n, γ) and $(n, n' \gamma)$ reactions in the target

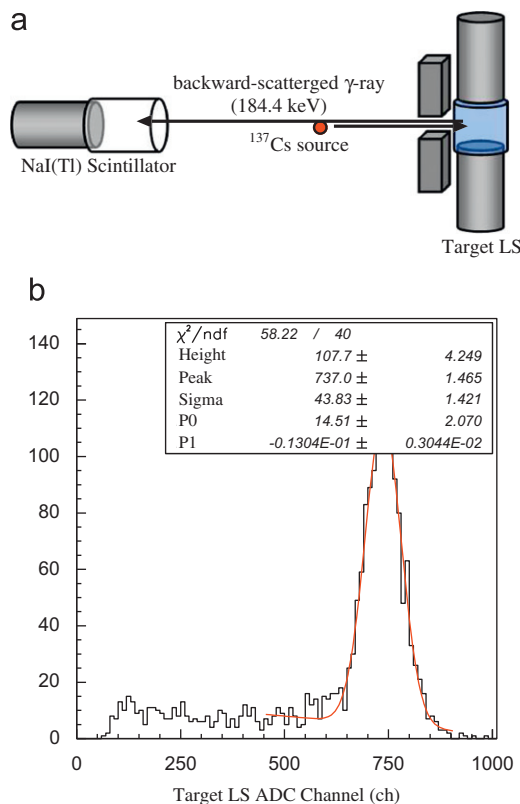


Fig. 2. Energy calibration of the target LS. (a) Experimental setup. (b) Energy spectrum for the coincidence events between the target LS and NaI(Tl) detector.

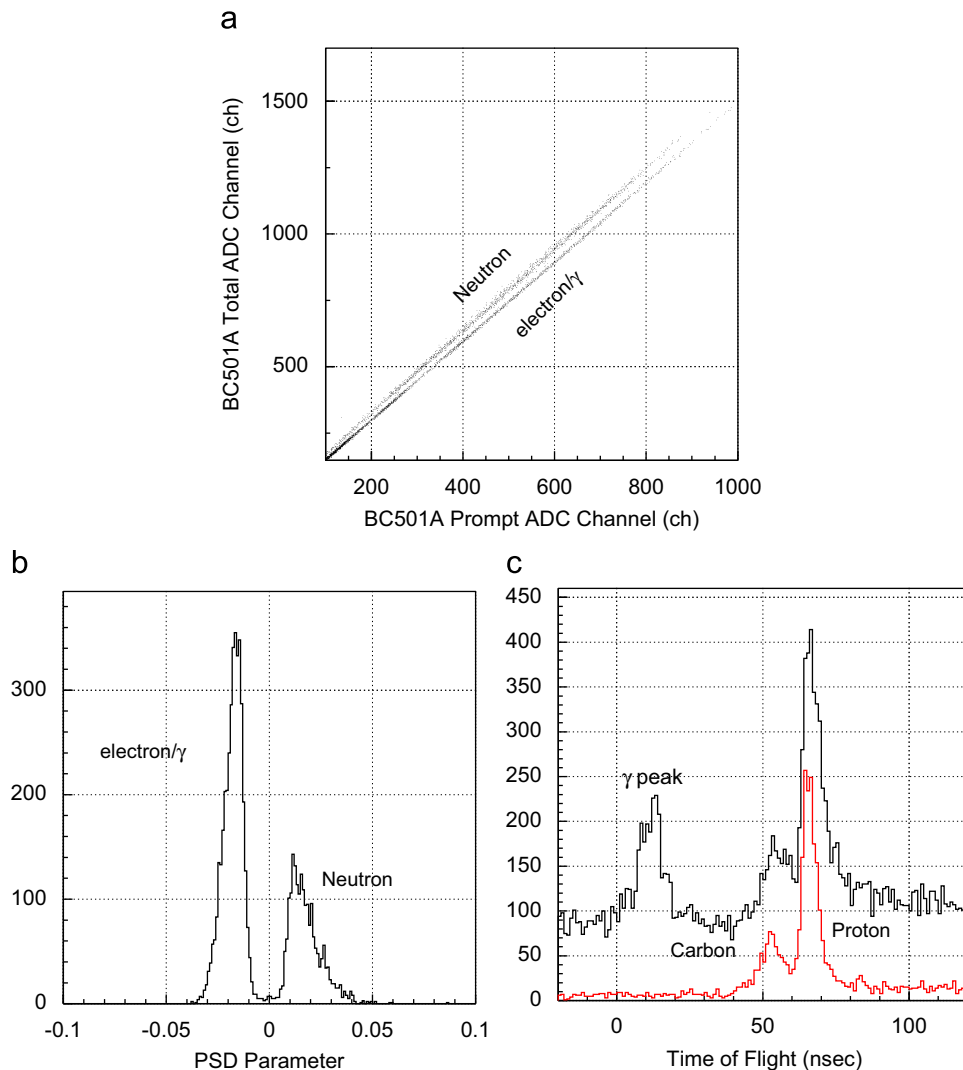


Fig. 3. Illustration of the event selection. The spectra were obtained with the coincident events with the BC501A detector placed at the scattering angle of 30° . (a) Two-dimensional plot between the total charge and the prompt charge for the BC501A detector. (b) Distribution of the PSD parameter for coincidence events. The ratio a in Eq. (3) works as an adjustable constant to set the n/γ separation to 0 on the PSD parameter. (c) TOF spectra for coincidence events observed under the higher PMT gain condition. The upper (lower) histogram represents TOF spectrum without (with) PSD selection in the BC501A detector. The TOF for γ -rays, neutrons scattered by protons and ^{12}C nuclei in the target LS are expected to be 9.2, 53.6, and 61.2 ns, respectively.

LS were reduced by PSD and TOF event selections. Backgrounds due to the accidental coincidence were also suppressed by the thick concrete wall. The concrete wall shielded neutrons and γ -rays directly from the TiT target and its surroundings, hence reduced the event rate in the BC501A detectors.

The peaks in the recoil energy spectra were fitted with Gaussian functions. The obtained Gaussian peak width was consistent with the expectation from superimposing the energy resolution of the target LS, the energy spread of the incident neutrons, and the scattering angle spread due to the size of the target and BC501A detectors.

Fig. 5(a)–(f) shows the recoil energy spectra obtained by applying TOF selection corresponding to the carbon recoils and subtracting the accidental background estimated in the off-time window. The spectrum of coincidence events with the BC501A placed at 10° is not shown, since almost all of observed events are considered to be induced by proton recoils as shown in Fig. 4(a).

The number of events in the low energy region in Fig. 5 was corrected by the detection (trigger) efficiency estimated with the Monte Carlo simulation. The trigger was made from a coincidence

of two PMT signals within a time window of 30 ns. The efficiency to detect single photoelectrons was estimated to be 75%. The number of observable photoelectrons followed a Poisson distribution with a mean photoelectron yield of 0.6 photoelectrons/keV. The trigger efficiency was evaluated using these conditions.

The peaks in the energy spectra induced by carbon recoils were fitted with Gaussian functions superimposed on a flat background. The peak width was fixed with the expected value based on the detector energy resolution, the width of the PMT photoelectron signal, and the recoil energy spread due to the size of the detectors. The peak in Fig. 5(a) spectrum was considered to be located below the energy threshold (~ 4 keV), so that the QF for carbon recoils at 80.2 keV was evaluated as an upper limit.

The QF for proton and carbon recoils was evaluated as the ratio of the observed electron-equivalent energy to the calculated recoil energy, according to Eq. (2). The QF obtained for each recoil energy are listed in Table 1. Fig. 6(a) and (b) shows the energy dependence of the QF for the protons and the carbon nuclei, respectively, in the KamLAND LS.

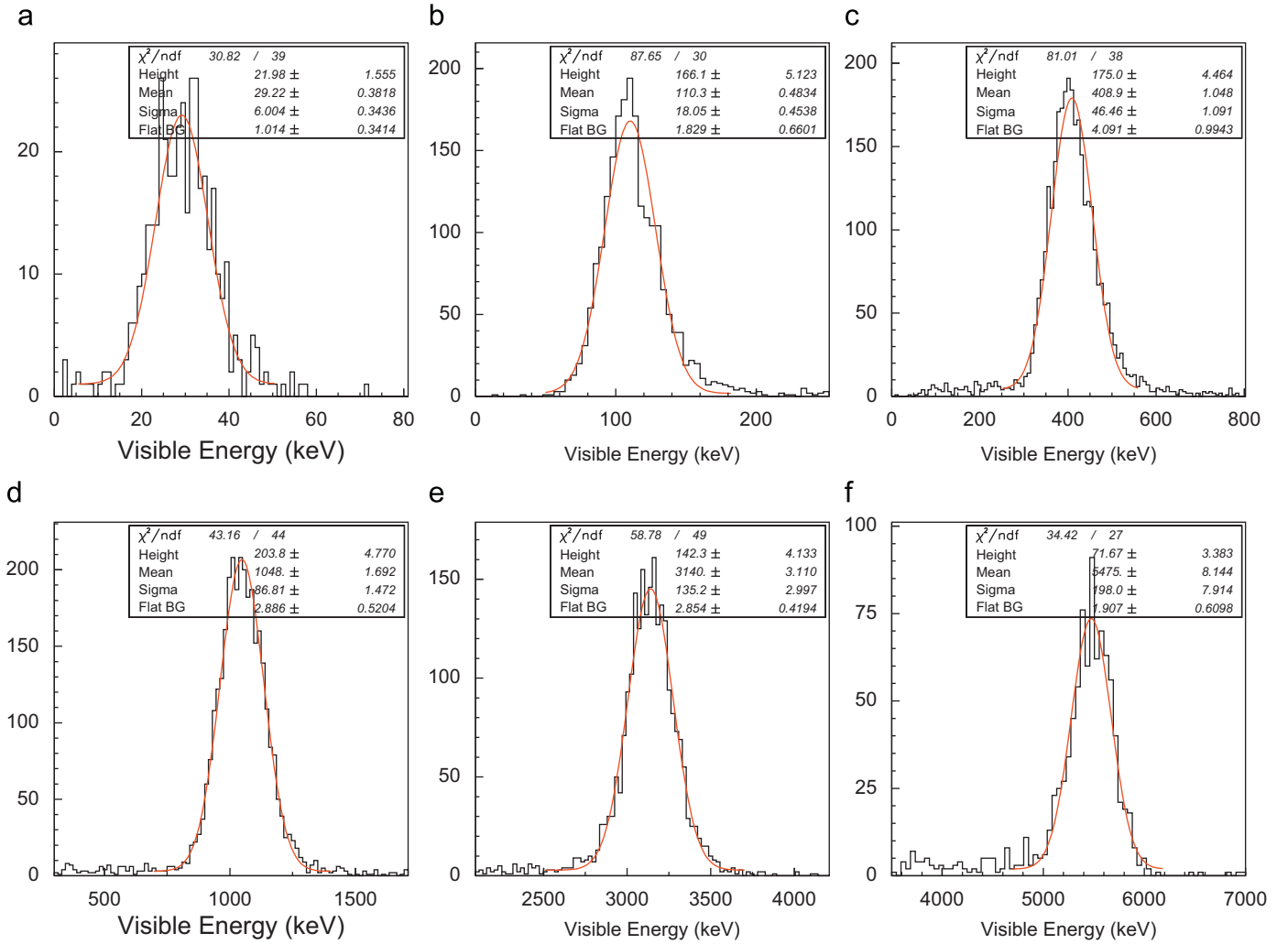


Fig. 4. Energy spectra for proton recoils in the target LS coincident with the BC501A detectors placed at angles of (a) 10°, (b) 15°, (c) 22°, (d) 30°, (e) 45°, and (f) 60°. The X-axis represents the electron equivalent energy of the target LS calibrated with backward-scattered γ -rays from ^{137}Cs . The spectrum in (a) alone was observed under the higher PMT gain condition, while the other spectra were recorded under the lower gain condition. Selected TOF range for each spectrum corresponded with the scattered neutron energy by protons in the target LS.

3.3. Uncertainties

The error bars for both axes in Fig. 6 include not only the statistical, but also systematic errors. The systematic error sources were evaluated as follows:

- (1) Uncertainty of mean recoil energy
 - (a) error of the incident neutron beam energy (0.33%);
 - (b) uncertainties of the deployment positions of detectors (5 mm for three forward angles and 10 mm for four backward angles).
- (2) Calibration error of the electron-equivalent energy scale for the target detector
 - (a) error of the peak fitting for the calibration spectrum (0.20% and 0.21% for low and high gain mode, respectively);
 - (b) gain shift during the experiment (0.30%);
 - (c) uncertainty of detector placement for the calibration setup (0.13%);
 - (d) baseline shift of PMT signal during measurement (0.30 keV in electron-equivalent scale)
- (3) Error of the detection efficiency estimated for the low energy region (0.25–0.30 keV for peak fitting results of carbon recoil spectra).

The errors of QF in Table 1 include the contribution of errors in the mean recoil energy.

4. Discussion

The measured energy dependence of the proton QF was fitted with the Birks formula Eq. (1) taking kB as a free parameter. The energy loss dE/dx values for protons and electrons were given by SRIM2008 [20] code. The differential light output was integrated along every step to calculate the QF of the proton for each energy. In this procedure, a linear light output response for electrons was assumed, and the light output of 477.3 keV was used to normalize.⁷ The dotted curve in Fig. 6(a) represents the calculated QF for proton recoils using the Birks formula Eq. (1) with the best-fit parameter $kB = 1.34 \times 10^{-2} \text{ g/cm}^2/\text{MeV}$. There is the significant difference between the calculated QF and the experimental data obtained in the present study.

⁷ In the KamLAND analysis, the energy scale is normalized by the observed light yield of 2.2 MeV γ -rays. The QF for 477.3 keV electrons is estimated to be 0.982. The result in the current work is applied to the KamLAND analysis with renormalization by this factor.

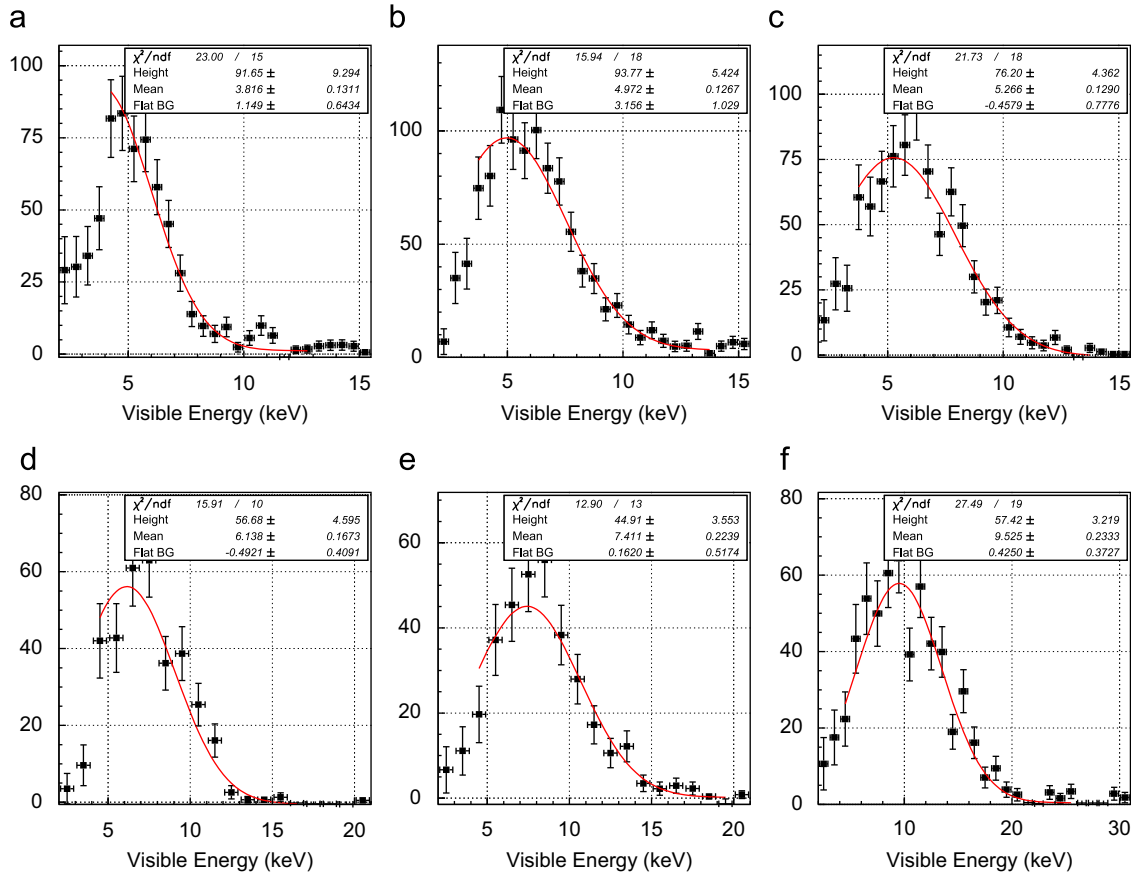


Fig. 5. Energy spectra for carbon recoils in the target LS coincident with the BC501A detectors placed at angles of (a) 15°, (b) 22°, (c) 30°, (d) 45°, (e) 60°, and (f) 90°. The representation of the X-axis is the same as in Fig. 4. All energy spectra were obtained under the higher gain condition with background subtraction and trigger efficiency correction. The best-fit Gaussian functions with the fixed width are also shown.

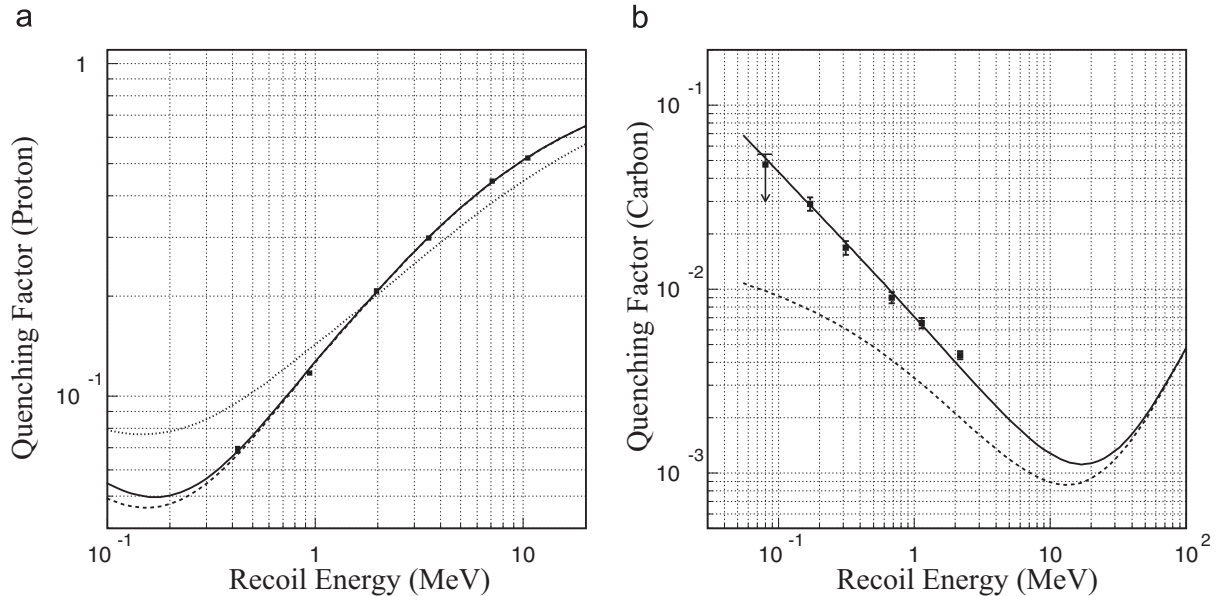


Fig. 6. The measured light output responses for proton and carbon recoils in the KamLAND LS. (a) The energy dependence of the QF for proton recoils. The dotted curve represents the Birks formula of Eq. (1) with the best-fit parameter $kB = 1.34 \times 10^{-2} \text{ g/cm}^2/\text{MeV}$. The dashed line shows the best-fit function using Eq. (4) with $kB = 7.79 \times 10^{-3} \text{ g/cm}^2/\text{MeV}$ and $C = 1.64 \times 10^{-5} (\text{g/cm}^2/\text{MeV})^2$. The solid line shows the best-fit function using Eq. (5) with the best-fit parameters given by the carbon data, $S_p/S_n = 0.371$. (b) The energy dependence of the QF for carbon recoils. The plot of the lowest recoil energy represents the upper limit of QF. The solid line shows the best-fit function using Eq. (5). The dashed line shows the curve calculated using Eq. (4), the Birks formula without the correction of the nuclear energy loss terms but with the same parameters as proton recoils.

Chou [13] proposed an empirical extension of the Birks formula to better reproduce the observed light output at low energies,

$$\frac{dL}{dx} = \frac{S \left(\frac{dE}{dx} \right)}{1 + kB \left(\frac{dE}{dx} \right) + C \left(\frac{dE}{dx} \right)^2} \quad (4)$$

where C is an adjustable constant for higher-order correction. The measured energy dependence of the proton QF was fitted with the extended Birks formula taking kB and C as free parameters. The obtained results were $kB = (7.79 \pm 0.13) \times 10^{-3} \text{ g/cm}^2/\text{MeV}$ and $C = (1.64 \pm 0.05) \times 10^{-5} (\text{g/cm}^2/\text{MeV})^2$ ($\chi^2/\text{n.d.f.} = 10.3/4$). The best-fit response curve is shown in Fig. 6(a) (dashed line). The observed QF for proton recoils is well described by the extended Birks formula within the measured energy range.

The QF obtained for carbon recoils was also fitted with the extended Birks formula. However, we could not find parameters in a reasonable range that accounted for the observed energy dependence of QF. The dashed curve in Fig. 6(b) represents the calculated QF for carbon recoil using the extended Birks formula Eq. (4) with the same parameters for proton recoil, $kB = 7.79 \times 10^{-3}$ and $C = 1.64 \times 10^{-5}$. The extended Birks formula does not describe the steep increase in the QF for carbon recoils as the recoil energy decreases below a few hundred keV.

A similar discrepancy was reported in Ref. [21], where it was pointed out that the energy loss was attributed to two terms: nuclear energy loss and electronic energy loss. These two terms contribute differently to the quenching (scintillating) processes. The authors proposed the generalized Birks formula to account for the steep increase of QF at low energy region,

$$\frac{dL}{dx} = \frac{S_e \left(\frac{dE}{dx} \right)_e + S_n \left(\frac{dE}{dx} \right)_n}{1 + kB_e \left(\frac{dE}{dx} \right)_e + kB_n \left(\frac{dE}{dx} \right)_n}$$

with the constraint $kB_e/kB_n = 3250$. Here we extend this generalized Birks formula by introducing the higher-order correction as Chou's extension in Eq. (4):

$$\frac{dL}{dx} = \frac{S_e \left(\frac{dE}{dx} \right)_e + S_n \left(\frac{dE}{dx} \right)_n}{1 + kB_e \left(\frac{dE}{dx} \right)_e + C_e \left(\frac{dE}{dx} \right)_e^2 + kB_n \left(\frac{dE}{dx} \right)_n} \quad (5)$$

We fitted the experimental data with the above formula with the same parameters as for protons, $kB_e = 7.79 \times 10^{-3}$ and $C_e = 1.64 \times 10^{-5}$, and the same constraint $kB_e/kB_n = 3250$ as in Ref. [21]. The only free parameter was S_e/S_n . The obtained result was $S_e/S_n = 0.371 \pm 0.033$ ($\chi^2/\text{n.d.f.} = 9.45/4$). The calculated curve with the best-fit parameter is shown in Fig. 6(b) (solid line). The energy dependence of carbon QF is well described by generalizing Birks formula with the extension of the nuclear energy loss term.

The nuclear energy loss term in Eq. (5) does not largely affect the response curve for the proton, as shown in Fig. 6(a) (solid line), since the fraction of the nuclear energy loss for protons is relatively small and negligible in the measured recoil energy range, compared to that for carbon. The generalized Birks formula of Eq. (5) successfully gives a simultaneous description of the light output responses for both protons and carbon nuclei in the KamLAND LS with the same values of the parameters. It is reasonably assumed that the light output response can be deduced for any other charged particles by replacing the energy loss dE/dx table in Eq. (5). However, the calculated QF for α -particles is almost as half as the measured one in the KamLAND detector. Therefore data for different types of charged particles and different energy range are required to investigate

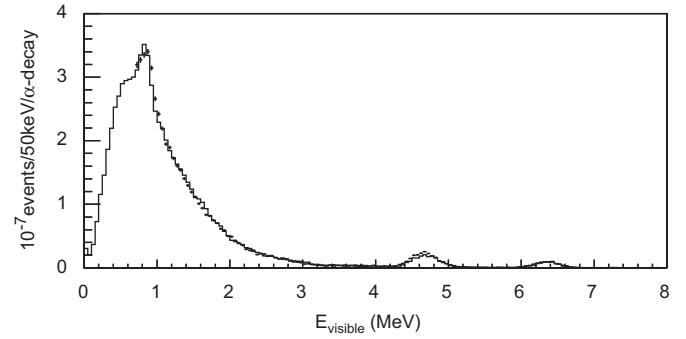


Fig. 7. The energy spectrum of the ^{13}C - α calibration source measured in the KamLAND detector. The plots correspond to experimental data, and the histogram shows the simulated spectrum by the Monte Carlo using the measured light output responses for nuclear recoils.

universal description of the light output responses in further study.

The reliability of the obtained response curves was demonstrated by the calibration data in the KamLAND experiment. The ^{13}C - α calibration source was newly developed [22], especially to estimate the branching fraction of the $^{13}\text{C}(\alpha, n)^{16}\text{O}^{(*)}$ reaction to the excited states of ^{16}O . The calibration spectrum of the ^{13}C - α source was well reproduced by the Monte Carlo simulation in which the measured light output responses are implemented,⁸ as shown in Fig. 7. The result of the proton QF in the present work was also in good agreement with that obtained in the different measurement [24] for the KamLAND LS.

The systematic uncertainty of the energy scale in the expected $^{13}\text{C}(\alpha, n)^{16}\text{O}^{(*)}$ background spectrum was reduced to be 1%, while the energy scale error of 10% was assigned to the previous analysis in Ref. [4]. The measured light output responses are allowed to improve the reliability for the anti-neutrino analysis in the KamLAND [3].

5. Conclusion

The precise measurement of the light output response in the KamLAND LS was performed using a monochromatic neutron beam. Both proton and carbon elastic scatterings were induced by the neutrons in the KamLAND LS target, which also functioned as a detector to measure the scintillation light yield. The obtained recoil energy spectra were free from backgrounds under the detector setup in the present study.

The observed energy dependence of QF for proton recoils is well described by the Birks formula with a higher-order correction. The obtained best-fit parameters were $kB = 7.79 \times 10^{-3} \text{ g/cm}^2/\text{MeV}$ and $C = 1.64 \times 10^{-5} (\text{g/cm}^2/\text{MeV})^2$. The total systematic errors for the proton recoil measurement were reduced to be 1–2%.

The light output response for carbon recoils was relatively small. The signals induced by the elastic scattering with carbon nuclei were distributed below 10 keV for the measured recoil energy range. The observed energy dependence of QF for carbon recoils was also well described by modifying the Birks formula with the nuclear energy loss terms kB_n and S_n .

The obtained light output responses in this study were used to estimate the background energy spectrum of the $^{13}\text{C}(\alpha, n)^{16}\text{O}$ reaction in the KamLAND. The obtained results play an important

⁸ An approximate function for directly converting the proton recoil energy into the energy scale in the KamLAND analysis was used in the simulation as $E_{\text{vis}} = a_1(1 - \exp(a_2 + a_3 E_p))E_p$, the similar function proposed by Madey et al. [23], with $a_1 = 0.581$, $a_2 = -3.35 \times 10^{-2}$, and $a_3 = -0.207$. This direct conversion function is independent of the code of dE/dx tables.

role in improving our understanding of the background spectrum in the current KamLAND analysis.

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